

Nova Corps Scramble

MIRAGE 2.0 — Binary Exploitation — Official Writeup

CATEGORY	DIFFICULTY	FILES
Binary Exploitation (Logic / Integer Overflow)	Medium	nova_corps_scramble
DEPLOYMENT	VULNERABILITY	STATUS
Offline binary — run locally	Unsigned 32-bit integer overflow	Solved

Challenge Premise

The binary presents a “Nova Corps Resource Terminal” tracking two 32-bit unsigned counters, Energy (starts at 100) and Decoys (starts at 0). The stated objective is to “provide the correct energy requirement to generate the Master Override.”

Reversing the Binary

The menu offers three options:

- **[1] Allocate Decoys** — costs amount $\times 10$ Energy, gains amount Decoys.
- **[2] Recover Decoys** — spends amount Decoys, gains amount Energy (1:1).
- **[3] Generate Master Override** — checks whether Energy is above a threshold; if so, prints the flag.

Disassembly of the Energy check shows an unsigned 32-bit comparison against 0xF423F (999,999):

```
cmp    dword [energy], 0xF423F
jbe    reject          ; unsigned compare – Energy must be > 999999
```

Critically, every arithmetic operation — amount $\times 10$ for the allocation cost, and the additions/subtractions for both operations — is carried out in plain 32-bit registers with no overflow checking. User input is read via `strtoul` into a 32-bit value, so any value above $2^{32}-1$ silently truncates to its low 32 bits.

Exploitation

Choosing amount = 2147483648 (0x80000000) for option 1 makes the cost calculation $10 \times 2147483648 \bmod 2^{32} = 0$ — the allocation is effectively free ($0 \leq \text{Energy}=100$), while Decoys jumps to 2,147,483,648. Feeding that exact same value back into option 2 then adds 2,147,483,648 to Energy, producing $100 + 2147483648 = 2147483748$ — far past the 1,000,000 threshold — via a deliberate unsigned integer overflow rather than any legitimate accumulation.

Participant Solve Steps

```
1
2147483648
2
2147483648
3
```

Sent in order to the binary's prompts, this sequence allocates 2,147,483,648 decoys at zero net cost, recovers them for the same amount of Energy, and then requests the Master Override with Energy far above the threshold.

FLAG

```
MIRAGE{m4th_1s_h4rd_wh3n_b0unds_4r3_f1x3d}
```

Verification

The flag string is stored XOR-obfuscated (key 0x55) inside the binary's `run_terminal` routine; it was decoded statically and then confirmed by actually running the input sequence above against the live binary, which printed the identical flag.